# Reel Returns API 1.0.0 OAS 3.1

/openapi.json

Welcome to the **Reel Returns API**! This service bridges the gap between historical cinema and modern economics by calculating what the box office revenue of older movies would be in today's economy using Consumer Price Index (CPI) data.

## Core Features & Endpoints

- **Public Movie Data:** Search the vault for films and retrieve their inflation-adjusted budgets, revenues, and Return on Investment (ROI).
- **Analytics Engine:** View the all-time profitability leaderboard, or use the *Success Predictor* to simulate the financial risk of a proposed movie based on historical genre and budget comparables.
- **Admin Vault:** Protected CRUD operations to manage the movie database and update annual inflation rates.

## Authentication Process

Endpoints tagged with **(Admin)** are protected and require authentication to access.

To authenticate:

1. Click the **Authorize** button (the padlock icon) at the top right of this page or next to any protected endpoint.
2. Enter your valid Admin API Key into the `X-API-Key` value box.
3. Click **Authorize** to lock it in. All subsequent requests to Admin endpoints will now automatically include your key.

Authorize

## Root

GET / API Health Check & Welcome

Returns a welcome message and confirms the API is successfully running.

**Parameters** Try it out

No parameters

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Connection<br>Media type<br>application/json<br>Controls Accept header.<br>**Example Value** \| Schema<br>`{ "message": "Welcome to the Reel Returns API! Go to /docs to see the documentation." }` | *No links* |

# Movies

GET `/movies/adjusted/{movie_id}`
Get inflation-adjusted financials for a specific movie

Retrieves a movie by ID, fetches historical and modern CPI data, and calculates the inflation-adjusted budget, revenue, and Return on Investment (ROI). Defaults to a 1.0 multiplier if future inflation data is missing.

**Parameters** Try it out

| Name | Description |
|---|---|
| **movie_id** * required<br>`integer`<br>*(path)* | movie_id |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Response<br><br>Media type<br>application/json<br>Controls Accept header.<br><br>**Example Value** \| Schema | *No links* |

```
{
  "movie_id": 1,
  "title": "Titanic",
  "release_year": 1997,
  "original_budget": 200000000,
  "original_revenue": 2187463944,
  "adjusted_budget": 380000000,
  "adjusted_revenue": 4156000000,
  "roi_percentage": 993.73,
  "genres": "Drama, Romance"
}
```

| | | |
|---|---|---|
| 404 | Movie not found<br><br>Media type<br>application/json<br><br>**Example Value** | *No links* |

```
{
  "detail": "Movie not found in the database"
}
```

| | | |
|---|---|---|
| 422 | Validation Error (e.g., providing a string instead of an integer for movie_id)<br><br>Media type<br>application/json<br><br>**Example Value** | *No links* |

```
{
  "detail": [
    {
      "loc": [
        "path",
        "movie_id"
      ],
      "msg": "value is not a valid integer",
      "type": "type_error.integer"
    }
  ]
}
```

GET /movies/search/ Search for movies by title

Performs a case-insensitive search for movies matching the title query. Supports pagination via skip and limit parameters.

**Parameters** Try it out

| Name | Description |
| --- | --- |
| **title** * required<br>`string`<br>*(query)* | title |
| skip<br>`integer`<br>*(query)* | *Default value* : 0<br>0 |
| limit<br>`integer`<br>*(query)* | *Default value* : 10<br>10 |

**Responses**

| Code | Description | Links |
| --- | --- | --- |
| 200 | Successful Search<br><br>Media type<br>application/json<br>Controls Accept header.<br><br>**Example Value** \| Schema | *No links* |

```
{
  "matches_returned": 1,
  "skip": 0,
  "limit": 10,
  "results": [
    {
      "movie_id": 42,
      "title": "Avatar",
```

| Code | Description | Links |
|---|---|---|

```
      "release_year": 2009,
      "genres": "Action, Science Fiction"
    }
  ]
}
```

| | | |
|---|---|---|
| 404 | No matches found | *No links* |

Media type

**application/json**

**Example Value**

```
{
  "detail": "No movies found matching 'query'."
}
```

| | | |
|---|---|---|
| 422 | Validation Error | *No links* |

## Analytics

**GET** `/analytics/success-predictor/`
Predict success based on proposed budget and genre

Calculates historical success rates, genre averages, and top comparable movies by defining a cohort bracket (+/- 20% of the proposed budget).

**Parameters** **Try it out**

| Name | Description |
|---|---|
| **proposed_budget** * required<br>`number`<br>*(query)* | proposed_budget |
| **genre** * required<br>`string`<br>*(query)* | genre |

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successful Analysis<br><br>Media type<br>application/json<br>Controls Accept header.<br><br>**Example Value** \| Schema | *No links* |

```
{
  "predictor_parameters": {
    "proposed_budget": 100000000,
    "genre": "Action"
  },
  "risk_assessment": {
    "cohort_size": 25,
    "success_rate_percentage": 72.5,
    "rating": "Strong Green Light",
    "colour_code": "green",
    "analysis": "Low Risk."
  },
  "genre_insights": {
    "total_movies_analysed": 150,
    "average_historical_roi_percentage": 115.2,
    "closest_budget_comps": []
  }
}
```

| | | |
|---|---|---|
| 400 | Not Enough Data<br><br>Media type<br>application/json<br><br>**Example Value** | *No links* |

```
{
  "detail": "Not enough historical data for the 'Unknown'
genre in this budget bracket."
}
```

| | | |
|---|---|---|
| 422 | Validation Error (e.g., invalid budget format) | *No links* |

**GET** `/analytics/leaderboard` Get profitability leaderboard

Calculates and returns the most profitable movies of all time, adjusted for inflation. Sorted by ROI percentage in descending order.

## Parameters

Try it out

| Name | Description |
| --- | --- |
| top<br>integer<br>*(query)* | *Default value* : 10<br>10 |

## Responses

| Code | Description | Links |
| --- | --- | --- |
| 200 | Successful Leaderboard Generation<br><br>Media type<br>application/json<br>Controls Accept header.<br><br>**Example Value** \| Schema | *No links* |

```
{
  "leaderboard_size": 10,
  "top_movies": [
    {
      "movie_id": 1,
      "title": "Titanic",
      "release_year": 1997,
      "roi_percentage": 993.73
    }
  ]
}
```

| | | |
| --- | --- | --- |
| 400 | Database Error<br><br>Media type<br>application/json<br><br>**Example Value** | *No links* |

```
{
  "detail": "Not enough data to calculate a leaderboard."
}
```

| | | |
| --- | --- | --- |
| 422 | Validation Error | *No links* |

# Authentication

GET `/auth/verify` Verify Admin API Key

Tests the validity of the provided API key. Use this endpoint to confirm your authentication status before attempting to access protected Admin CRUD routes.

**Parameters** Try it out

No parameters

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successfully Authenticated<br>Media type<br>application/json<br>Controls Accept header.<br>**Example Value** \| Schema | *No links* |
| 403 | Invalid or Missing API Key<br>Media type<br>application/json<br>**Example Value** | *No links* |

```
{
  "status": "authenticated",
  "message": "API Key is valid"
}
```

```
{
  "detail": "Could not validate credentials"
}
```

# Movie Management (Admin)

POST /movies/ Create a new movie record

Adds a new movie to the database. Requires Admin API Key. Validates against duplicate titles released in the same year.

**Parameters** Try it out

No parameters

**Request body** required application/json

**Example Value** | Schema

```
{
  "budget": 350000000,
  "genres": "Action, Science Fiction",
  "release_year": 2022,
  "revenue": 2320000000,
  "title": "Avatar: The Way of Water"
}
```

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successfully Created<br>Media type<br>application/json<br>Controls Accept header.<br>**Example Value** \| Schema | *No links* |
| 400 | Duplicate Movie Error<br>Media type<br>application/json | *No links* |

```
{
  "message": "Added 'New Movie' to the database successful
ly.",
  "movie_id": 99
}
```

| Code | Description | Links |
| --- | --- | --- |
| | Example Value<br>`{ "detail": "Error: 'Movie Title' (2024) already exists in the database with ID 5." }` | |
| 403 | Not Authenticated | *No links* |
| 422 | Validation Error (e.g. negative budget provided) | *No links* |

**PUT** `/movies/{movie_id}` Update an entire movie record

Overwrites an existing movie record. Requires Admin API Key.

**Parameters** Try it out

| Name | Description |
| --- | --- |
| **movie_id** * required<br>`integer`<br>*(path)* | movie_id |

**Request body** required application/json

Example Value | Schema

```
{
  "budget": 350000000,
  "genres": "Action, Science Fiction",
  "release_year": 2022,
  "revenue": 2320000000,
  "title": "Avatar: The Way of Water"
}
```

**Responses**

| Code | Description | Links |
|---|---|---|
| 200 | Successfully Updated<br>Media type<br>application/json<br>Controls Accept header.<br>Example Value \| Schema | *No links* |
| 403 | Not Authenticated | *No links* |
| 404 | Movie Not Found | *No links* |
| 422 | Validation Error | *No links* |

```
{
  "message": "Successfully updated movie ID 99",
  "title": "Updated Title"
}
```

**PATCH** `/movies/{movie_id}` Partially update a movie record

Updates only the fields provided in the request payload without overwriting the entire record. Requires Admin API Key.

**Parameters** Try it out

| Name | Description |
|---|---|
| **movie_id** * required<br>`integer`<br>*(path)* | movie_id |

**Request body** required application/json

Example Value | Schema

```
{
  "title": "string",
  "release_year": 1888,
  "budget": 0,
  "revenue": 0,
  "genres": "string"
```

```
}
```

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successfully Patched<br>Media type<br>application/json<br>Controls Accept header.<br>Example Value \| Schema<br>`{ "message": "Successfully updated 'Title'!", "current_movie_state": {} }` | *No links* |
| 403 | Not Authenticated | *No links* |
| 404 | Movie Not Found | *No links* |
| 422 | Validation Error | *No links* |

**DELETE** `/movies/{movie_id}` Delete a movie

Permanently removes a movie from the database. Requires Admin API Key.

## Parameters

Try it out

| Name | Description |
|---|---|
| **movie_id** * required<br>`integer`<br>*(path)* | movie_id |

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successfully Deleted<br>Media type<br>application/json<br>Controls Accept header.<br>Example Value \| Schema<br>`{ "message": "Successfully deleted 'Movie Title' from the database." }` | *No links* |
| 403 | Not Authenticated | *No links* |
| 404 | Movie Not Found | *No links* |
| 422 | Validation Error | *No links* |

## Inflation Management (Admin)

**POST** `/inflation/` Add new inflation data

Creates a new CPI record for a specific year. Fails if the year already exists. Requires Admin API Key.

**Parameters** Try it out

No parameters

**Request body** required application/json

Example Value | Schema

```
{
  "cpi": 311.5,
  "year": 2024
}
```

## Responses

| Code | Description | Links |
|---|---|---|
| 200 | Successfully Created<br>Media type<br>application/json<br>Controls Accept header.<br>Example Value \| Schema<br>`{ "message": "Added new inflation data for 2024" }` | *No links* |
| 400 | Data Already Exists<br>Media type<br>application/json<br>Example Value<br>`{ "detail": "Inflation data for 2024 already exists. Please use the PUT endpoint to update it." }` | *No links* |
| 403 | Not Authenticated | *No links* |
| 422 | Validation Error | *No links* |

**PUT** `/inflation/{year}` Update existing inflation data

Updates the CPI value for an existing year. Fails if the year does not exist in the database. Requires Admin API Key.

### Parameters

Try it out

| Name | Description |
|---|---|
| **year** * required<br>`integer`<br>*(path)* | year |

| Name | Description |
| --- | --- |
| **cpi** * required<br>number<br>*(query)* | cpi |

## Responses

| Code | Description | Links |
| --- | --- | --- |
| 200 | Successfully Updated<br><br>Media type<br>application/json<br>Controls Accept header.<br><br>**Example Value** \| Schema<br><br>`{ "message": "Updated CPI for 2024 to 311.5" }` | *No links* |
| 403 | Not Authenticated | *No links* |
| 404 | Data Not Found<br><br>Media type<br>application/json<br><br>**Example Value**<br><br>`{ "detail": "Inflation data for 2024 not found. Please use the POST endpoint to create it." }` | *No links* |
| 422 | Validation Error | *No links* |

## Schemas

**InflationCreate** Collapse all `object`

year* integer

cpi* number

Examples Collapse all array

#0 Collapse all object

**cpi**=311.5

**year**=2024

## MovieAdjustedResponse Collapse all object

movie_id* integer

title* string

release_year* integer

original_budget* number

original_revenue* number

adjusted_budget* number

adjusted_revenue* number

roi_percentage* number

genres Collapse all (string | null)

Any of Collapse all (string | null)

#0 string

#1 null

## MovieCreateUpdate Collapse all object

title* string [1, 250] characters

release_year* integer [1888, 2027]

budget* number ≥ 0

revenue* number ≥ 0

genres Collapse all (string | null)

Any of Collapse all (string | null)

#0 string

#1 null

Examples Collapse all array

#0 Collapse all object

**budget**=350000000

**genres**="Action, Science Fiction"

**release_year**=2022

**revenue**=2320000000

**title**="Avatar: The Way of Water"

## MovieUpdate ^ Collapse all object

title ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string [1, 250] characters

#1 null

release_year ^ Collapse all (integer | null)

Any of ^ Collapse all (integer | null)

#0 integer [1888, 2027]

#1 null

budget ^ Collapse all (number | null)

Any of ^ Collapse all (number | null)

#0 number ≥ 0

#1 null

revenue ^ Collapse all (number | null)

Any of ^ Collapse all (number | null)

#0 number ≥ 0

#1 null

genres ^ Collapse all (string | null)

Any of ^ Collapse all (string | null)

#0 string

#1 null